Braillix - Electronics BOM (HAVE on top, BUY at bottom)

> Written 2026-07-30. Companion to `docs/MASTER_BOM.md` (which also covers mechanical parts). Scope: **one working single-cell demo.** Multi-cell/production parts are listed separately at the end so they are not confused with what you need now.

The headline: you need NO chips, NO resistors, NO capacitors

Every discrete component the circuit would normally need is **already built into the modules you own**:

What you'd normally add	Why you don't need it
Motor flyback diodes	**Built into the ULN2003A.** The chip has internal freewheeling diodes on its COM pin - that is what COM is for.
Motor driver transistors	**That IS the ULN2003.** Seven Darlington pairs, 500mA each. You need four.
Button pull-up resistors	ESP32 has **internal** pull-ups. Firmware uses `INPUT_PULLUP`.
Hall sensor pull-up / comparator	**On the blue MH module already** (that's the second IC and the trimmer pot).
Voltage regulator + decoupling caps	**On the ESP32 DevKit already** (AMS1117 + caps).
Reverse-polarity protection	[!] **NOT present** in the circuit. Mitigated by measurement: the owned jack was tested and is correct. Measure any new connector before trusting it.

Total passives to buy for the demo: zero.

PART 1 - WHAT YOU ALREADY HAVE

Cross-check each of these physically. The "how to identify it" column is so you can confirm you actually have the right thing.

#	Component	How to identify it	Qty	Status
1	**ESP32 DevKit (USB-C)**	Black PCB ~51x28mm, silver square shield, USB-C at one end, 15 pins per side, BOOT + EN buttons	1	[OK] HAVE (?350)
2	**28BYJ-48 stepper motor**	Silver can ~28mm dia, blue plastic base, 5 coloured wires into a white plug	1	[OK] HAVE (?160 w/ driver)
3	**ULN2003 driver board**	Small **blue** PCB, 4 red LEDs, white 5-pin socket, black 16-pin chip, IN1-IN4 + +/- pins	1	[OK] HAVE - **this is your motor driver, no chip to buy**
4	**Hall sensor module (MH-Sensor-Series)**	Small **blue** PCB, tiny black 3-pin sensor on one edge, blue trimmer pot, 1-2 LEDs, pins marked **AO DO GND VCC**	1	[OK] HAVE (?50) - [!] **must use AO (analog)**, and the pocket won't fit it (see note)
5	**5V / 3A power adapter**	Wall plug, barrel connector on the lead, "5V 3A" on the label	1	[OK] HAVE (?160 w/ jack)
6	**Inline female DC pigtail jack**	Black cylindrical barrel socket with red/black wires; no thread or nut	1	[OK] HAVE for testing; [OK] **polarity MEASURED CORRECT 2026-08-01** (red = +). No thread/nut, so mounting is a CAD problem.
6b	**5.5x2.1mm female panel-mount jack**	Threaded neck plus retaining nut, rated >=5V/3A	1	[X] SELECT/BUY for the final pod; record exact drawing/link before lid CAD

#	Component	How to identify it	Qty	Status
7	**Connecting wire (2m)**	Loose hookup wire	2m	[OK] HAVE (?20)
8	**Dupont jumper wires**	Ribbon of coloured wires with plastic pin housings	some	[OK] HAVE - need ~8x M-F and ~5x M-M
9	**Neodymium magnets 8x1mm**	Small silver discs, strongly attract each other	10	[OK] HAVE (?135) - for docking, **not** for cam homing
10	**Tactile switches 6x6x5mm**	Tiny square black buttons, 4 legs	3	[OK] HAVE - not needed for the demo
11	**Digital multimeter**	Handheld meter with DC volts	1	[OK] HAVE (bought 2026-08-01) - jack polarity measured and CONFIRMED correct

Notes on things you already have

#6 - [OK] polarity is VERIFIED CORRECT (2026-08-01). Measured with the multimeter: **red = positive.** This retires the single largest risk to the ESP32. Mark the positive wire physically (nail polish, marker, tag) so the information cannot be lost at the bench. There is still no reverse-polarity protection in the circuit, so any *new* power connector must be measured the same way before it is trusted.

#4 - your whole Hall module does not fit the base plate. v7.5 rebuilt an underside pocket for the *bare* TO-92 sensor only; the blue PCB is still far too large. M11b measures 1.6mm and the current recess is exactly 1.6mm, so dry-fit it before gluing. **For the demo this does not matter** - the module sits on the breadboard. For final assembly, desolder the bare 3-pin sensor from the module and run three wires back to it.

PART 2 - WHAT YOU NEED TO BUY

2A. For the demo (buy this week)

#	Component	Spec	Why	Est. ?
1	**Soldering station, 60W, TEMPERATURE-CONTROLLED**	"936" type, ceramic heater, 2.4mm chisel tip	[!!] **NOT OWNED.** [!] Do NOT buy a plain ?200 pencil iron - uncontrolled heat burns the flux off and makes soldering feel impossible. See `ELECTRONICS_PLAN.md` Part 9.	~1,300
2	**Breadboard**	Half-size or full-size, 400-830 tie points	The whole Tier-0 demo is breadboard-based, and it's the safe way to bring up wiring before soldering.	~100
3	**USB-C cable (DATA)**	Must carry data, not charge-only	To flash the ESP32. Charge-only cables look identical and fail silently. **Test the one you have first - you may already be fine.**	~150
4	**Solder wire**	**0.8mm, 60/40 rosin core**	For soldering wires flat to the ULN2003. Not plumbing solder.	~150
5	**Wire stripper / cutter**	Small, for 22-26 AWG	Prepping wire ends.	~200

Demo subtotal: ~?1,900 (or ~?1,750 if your existing USB-C cable does data). The soldering station is the bulk of it and is the one genuinely missing tool.

2B. For the mechanism (buy when the resin parts arrive)

#	Component	Spec	Qty	Est. ?
6	**Micro compression springs**	**2.0mm OD**, ~0.3mm stainless wire, ~4mm free length. Buy an assortment kit. [!!] Pen springs (4mm) **cannot** work - braille rows are 2.6mm apart.	6 + spares	~500
7	**Homing magnet**	[!!] **3mm dia x 1mm thick** neodymium - **CHANGED from 3x2mm in v7.5.** The cam disc floor is only 2mm, so a 2mm magnet's pocket cut clean through it and punched a hole across three cam tracks. 1mm leaves 0.8mm of floor and couples just as well. Your 8x1mm ones are too wide.	1 (buy 10)	~120
8	**M2.5 x 25mm bolts**	Top plate -> standoffs -> box bosses	4	~60
9	**M4 x 10mm screws**	Motor mounting ears. **v7.5: nuts no longer needed** - the right-hand ear sits under the spinning cam with nowhere to put a nut, so the screws now thread directly into dia 3.3 pilot holes in the base plate. Self-tapping/thread-forming M4 preferred.	2	~30
10	**M2 x 8mm self-tap**	Pod lid. **NOT M2x6** - through a 4mm lid that leaves only 2mm of thread.	2	~40
11	**Superglue or 5-min epoxy**	Glues the resin dot insert into the PETG plate. Epoxy preferred - you get time to seat it square.	1	~80
12	**Digital calipers**	150mm digital	Already in hand; M5, M11b, and M21 recorded on 2026-07-31.	1

Mechanism subtotal: ~?1,430

2C. Optional / nice to have

Component	Why	Est. ?
Hot glue gun + sticks	* Strain relief on every joint. **Reworkable** - this is why it beats epoxy while the pin map can still change. See `ELECTRONICS_PLAN.md` Part 12.	~250
Neutral-cure RTV silicone	Optional upgrade over hot glue for vibration. [!!] Must say **neutral cure** - the vinegar-smelling acetic kind corrodes copper.	~150
Flux paste	Makes soldering to ULN2003 pins much easier for a beginner	~100
Helping-hands / third hand	Not really optional - you cannot hold iron, solder and two wires with two hands	~200
Brass wool tip cleaner	Better than the wet sponge; keeps the tip alive	~100
Heat-shrink tubing	Insulating soldered joints	~80
Tweezers	Handling 2mm springs and 1mm linkages - genuinely hard without	~80

Component	Why	Est. ?
Desoldering pump/wick	Fixing bridges; also for lifting the TO-92 off the hall module	~120
Small drill bits 1.5/1.7/2.5mm	FDM holes print 0.2-0.3mm undersize	~150
Electrical tape	-	~30

2D. Cost of each ADDITIONAL cell (the scaling claim, priced)

You are building two. This is what a third, fourth or eighth costs - no custom parts at any point. Full architecture in `ELECTRONICS_PLAN.md` Part 5.

Per extra cell	Est. ?
28BYJ-48 motor + ULN2003 board	~160
Hall sensor module	~50
MCP23017 expander (DIP-28) + 3 address jumpers	~90
Wire, connector, heat-shrink	~50
Total per cell	**~?350**

One-off, only when you go past 2 cells: 2x 4.7k? pull-ups (~?5, brain end only). **One-off, only past 16 cells:** TCA9548A I2C mux (~?120).

[!] Cells 1 and 2 run **direct-GPIO** and need no expander at all - but building both muscle boards with the expander anyway is what makes the N-cell claim demonstrable rather than theoretical.

Totals

	?
Already spent (incl. multimeter)	~1,435
Demo essentials (2A)	~1,900
Mechanism (2B)	~1,430
Optional (2C)	~960
Everything	**~5,725 of ?15,000**
each cell beyond the 2nd (2D)	*~350*

The breadboard demo needs NO soldering at all - only the breadboard and a data cable (~?250). The soldering station is required to *assemble* a cell, not to demonstrate one.

PART 3 - NOT NEEDED (so you don't buy it by mistake)

Item	Verdict
Any motor driver IC	[X] You own it. The chip on the blue board **is** the ULN2003A.
ATmega328P / custom muscle board	[X] Never. Multi-cell is solved by an **MCP23017 per brick** (DIP-28, hand-solderable, ~?80) - see `ELECTRONICS_PLAN.md` Part 5. TQFP-32 at 0.8mm pitch is beyond basic soldering and would need PCBA ordering.
Resistors	[X] None for one or two cells. *(From 3 cells: exactly 2x 4.7k? I2C pull-ups, **at the brain end only, once, ever** - repeating them per brick kills the bus.)*
Capacitors	[X] None. Both modules and the DevKit carry their own.
Flyback diodes	[X] Built into the ULN2003A (that's what COM does).
Level shifters	[X] Not needed - hall runs on 3V3, so GPIO34 never sees 5V.
2mm bearing balls	[X] Obsolete since v7.1 - the dot is printed as a dome.
Pogo pin connectors	?? Deferred. Spec when needed: 4-pin, 2.54mm pitch, spring-loaded, ~0.5A.

Item	Verdict
Crystal / oscillator	[X] ESP32 and the modules have their own.

If you ever DO build the custom muscle board

Not needed now, but for the record - it's `pcb/braillix_muscle_board.kicad_pcb`, 34x44mm:

Part	Package	Note
ATmega328P-AU	**TQFP-32, 7x7mm square**	0.8mm pitch - **not hand-solderable at your level**
ULN2003A	**SOIC-16 rectangle**	Same chip you already own, just SMD
16MHz crystal	HC49	
2x 22pF, 2x 100nF	0402	Tiny - 1.0 x 0.5mm
100µF/10V	SMD electrolytic	
10k? x2, 4.7k? x2	0402	
SS14 Schottky	SMA	?? *this* is the reverse-polarity protection the prototype lacks

Recommendation: order it fully assembled (PCBA) or don't build it. 0402 passives and a TQFP-32 are not a beginner job, and none of it is needed for a working demo.